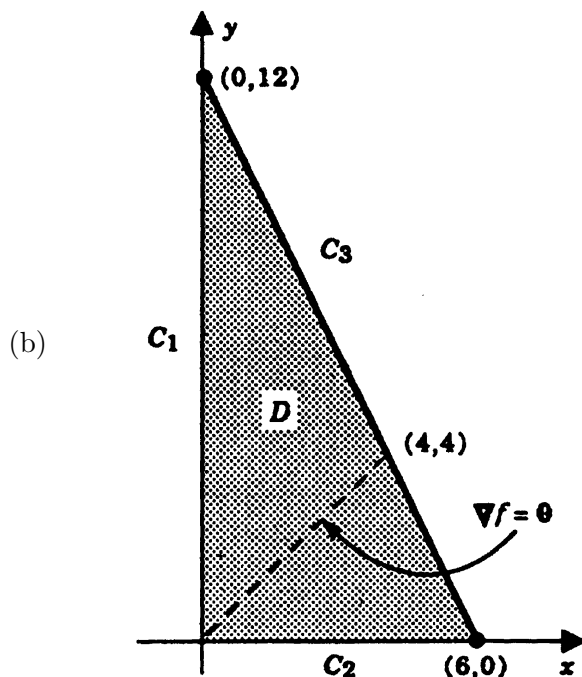


1. (a) $\nabla f = 2xy\mathbf{i} + x^2\mathbf{j}$ and $r'(t) = (e^t\mathbf{i} - e^{-t}\mathbf{j})$
 $\nabla f(r(t)) \cdot r'(t) = (2\mathbf{i} + e^{2t}\mathbf{j}) \cdot (e^t\mathbf{i} - e^{-t}\mathbf{j}) = e^t$
- (b) $\nabla f = \mathbf{i} - \mathbf{j}$ and $r'(t) = (a\mathbf{i} - ab\sin at\mathbf{j})$
 $\nabla f(r(t)) \cdot r'(t) = (\mathbf{i} - \mathbf{j}) \cdot (a\mathbf{i} - ab\sin at\mathbf{j}) = a(1 + b\sin at)$
- (c) $\nabla f = \frac{-2x}{1+(y^2-x^2)^2}\mathbf{i} + \frac{2y}{1+(y^2-x^2)^2}\mathbf{j} \Rightarrow \nabla f(r(t)) = \frac{-2\sin t}{1+\cos^2 2t}\mathbf{i} + \frac{2\cos t}{1+\cos^2 2t}\mathbf{j}$
and $r'(t) = \cos t\mathbf{i} - \sin t\mathbf{j}$
 $\nabla f(r(t)) \cdot r'(t) = \left(\frac{-2\sin t}{1+\cos^2 2t}\mathbf{i} + \frac{2\cos t}{1+\cos^2 2t}\mathbf{j} \right) \cdot (\cos t\mathbf{i} - \sin t\mathbf{j}) = \frac{-4\sin t \cos t}{1+\cos^2 2t} = \frac{-2\sin 2t}{1+\cos^2 2t}$
2. (a) $\nabla f = 2xe^{x^2-y^2}(1+x^2+y^2)\mathbf{i} + 2ye^{x^2-y^2}\mathbf{j} = \mathbf{0}$ at $(0,0), (0,1), (0,-1)$
 $A = \frac{\partial^2 f}{\partial x^2} = 2xe^{x^2-y^2}(2x+2x^3+2xy^2) + e^{x^2-y^2}(2+6x^2+2y^2)$
 $B = \frac{\partial^2 f}{\partial y \partial x} = -2ye^{x^2-y^2}(2x+2x^3+2xy^2) + e^{x^2-y^2}(4xy)$
 $C = \frac{\partial^2 f}{\partial y^2} = -2ye^{x^2-y^2}(2y-2yx^2-2y^3) + e^{x^2-y^2}(2-2x^2-6y^2)$
At $(0,0)$, $AC - B^2 = (2)(2) = 4 > 0$ and $A > 0 \Rightarrow (0,0)$ is the local minimum; the value is 0
At $(0, \pm 1)$, $AC - B^2 = (4e^{-1})(-4e^{-1}) = -16e^{-1} < 0 \Rightarrow (0, \pm 1)$ are saddle points
- (b) $\nabla f = (2x+2y+2)\mathbf{i} + (2x+6y+10)\mathbf{j} = \mathbf{0}$ only at $(1, -2)$
 $A = \frac{\partial^2 f}{\partial x^2} = 2, B = \frac{\partial^2 f}{\partial y \partial x} = 2, C = \frac{\partial^2 f}{\partial y^2} = 6$
 $\Rightarrow D = AC - B^2 = 2 \cdot 6 - 2^2 = 8 > 0$ and $A > 0$
Therefore $(1, -2)$ is the local minimum; the value is -8 .
- (c) $\nabla f = \left(\frac{1}{y} + \frac{y}{x^2} \right)\mathbf{i} + \left(-\frac{x}{y^2} - \frac{1}{x} \right)\mathbf{j} = \frac{x^2+y^2}{x^2y}\mathbf{i} = \frac{x^2+y^2}{xy^2}\mathbf{j}$ is never $\mathbf{0}$
Therefore, there is no any stationary points and local extreme values.
- (d) $\nabla f = -\sin x \cosh y\mathbf{i} + \cos x \sinh y\mathbf{j}$ at $(-\pi, 0), (0, 0), (\pi, 0)$
 $A = \frac{\partial^2 f}{\partial x^2} = -\cos x \cosh y, B = \frac{\partial^2 f}{\partial y \partial x} = -\sin x \sinh y, C = \frac{\partial^2 f}{\partial y^2} = \cos x \cosh y$
 $\Rightarrow D = AC - B^2 = -\cos^2 x \cosh^2 y - \sin^2 x \sinh^2 y < 0$
Therefore, $(-\pi, 0), (0, 0), (\pi, 0)$ are saddle points.
3. (a) $\nabla f = (2x+ky)\mathbf{i} + (2y+kx)\mathbf{j}$ and then $\nabla f(0,0) = \mathbf{0}$ is independent of the value k.
- (b) $f_{xx} = 2, f_{xy} = k, f_{yy} = 2 \Rightarrow D = 4 - k^2$
Thus, $D < 0$ for $|k| > 2$ and $(0,0)$ is a saddle point.
- (c) $D > 0$ for $|k| < 2$ and since $f_{xx} = 2 > 0$, $(0,0)$ is a local minimum.
- (d) The test is inconclusive when $D = 4 - k^2 = 0$, that is, for $k = \pm 2$.
(If $k = \pm 2$, $f(x,y) = (x \pm y)^2$ and $(0,0)$ is a minimum)
4. (a) $\nabla f = 2(x-3)\mathbf{i} + 2y\mathbf{j} = \mathbf{0}$ at $(3,0)$ which is not in the interior of D .
Boundary:
On $y = x^2$, $f = (x-3)^2 + x^4$, $\frac{df}{dx} = 2(x-3) + 4x^3 = 0$ at $x = 1 \Rightarrow (1,1)$
On $y = 4x$, $f = (x-3)^2 + 16x^2$, $\frac{df}{dx} = 2(x-3) + 32x = 0$ at $x = \frac{3}{17} \Rightarrow (\frac{3}{17}, \frac{12}{17})$

So the maximum and minimum occur at one or more of the following points:

$$(0,0), (4,16), (1,1), \left(\frac{3}{17}, \frac{12}{17}\right)$$

Evaluating f at these points, we find that $f(1,1) = 5$ is the absolute minimum of f ; $f(4,16) = 257$ is the absolute maximum of f .



$\nabla f = 2(x-y)\mathbf{i} + 2(x-y)\mathbf{j} = \mathbf{0}$ at each point of the line segment $y = x$ from $(0,0)$ to $(4,4)$.

Since $f(x,x) = 0$ and $f(x,y) \geq 0$, f takes on its minimum of 0 at each of these points. Now we consider the boundary of D , we parametrize each of the triangle:

$$C_1 : r_1(t) = t\mathbf{j}, \quad t \in [0, 12]$$

$$C_2 : r_2(t) = t\mathbf{i}, \quad t \in [0, 6]$$

$$C_3 : r_3(t) = t\mathbf{i} + (12-2t)\mathbf{j}, \quad t \in [0, 6]$$

and observe from

$$f(r_1(t)) = t^2, \quad t \in [0, 12] ;$$

$$f(r_2(t)) = t^2, \quad t \in [0, 6] ;$$

$$f(r_3(t)) = (3t-12)^2, \quad t \in [0, 6]$$

that f takes on its maximum of 144 at the point $(0,12)$.

(c) $\nabla f = (2x+1)\mathbf{i} + (8y-2)\mathbf{j} = \mathbf{0}$ at $(-\frac{1}{2}, \frac{1}{4})$ in D ; $f(-\frac{1}{2}, \frac{1}{4}) = -\frac{1}{2}$

Next we consider the boundary of D , we parametrize the ellipse by $C : r(t) = 2\cos t\mathbf{i} + \sin t\mathbf{j}, \quad t \in [0, 2\pi]$

The values of f on the boundary are given by the function

$$F(t) = f(r(t)) = 4\cos^2 t + 4\sin^2 t + 2\cos t - 2\sin t = 4 + 2\cos t - 2\sin t, \quad t \in [0, 2\pi]$$

$$F'(t) = f'(r(t)) = -2\sin t = 2\cos t : F'(t) = 0 \Rightarrow \cos t = -\sin t \Rightarrow t = \frac{3}{4}\pi, \frac{7}{4}\pi$$

Evaluating F at the endpoints and critical numbers, we have:

$$F(0) = F(2\pi) = f(2, 0) = 6; F(\frac{3}{4}\pi) = f(-\sqrt{2}, \frac{\sqrt{2}}{2}) = 4 - 2\sqrt{2};$$

$$F(\frac{7}{4}\pi) = f(\sqrt{2}, -\frac{\sqrt{2}}{2}) = 4 + 2\sqrt{2}$$

f takes on its absolute maximum of $4 + 2\sqrt{2}$ at $(\sqrt{2}, -\frac{\sqrt{2}}{2})$; f takes on its absolute minimum of $4 - 2\sqrt{2}$ at $(-\sqrt{2}, \frac{\sqrt{2}}{2})$.

5. $f(x, y) = x^2 + y^2 \Rightarrow \nabla f = 2x\mathbf{i} + 2y\mathbf{j}$ and $g(x, y) = xy - 1 \Rightarrow \nabla g = y\mathbf{i} + x\mathbf{j}$

So $\nabla f = \lambda \nabla g \Rightarrow 2x = \lambda y$ and $2y = \lambda x$

Multiplying the first equation by x and the second equation by y , we get $2x^2 = \lambda xy = 2y^2$

Thus, $x = \pm y$. From $g(x, y) = 0$ we conclude that $x = y = \pm 1$. The points $(1, 1)$ and $(-1, -1)$ clearly give a minimum, since f represents the square of the distance of a point on the hyperbola from the origin. The minimum is 2.

6. Since f is continuous and the ellipse is closed and bounded, the maximum exists.

$$f(x, y) = xy^2 \Rightarrow \nabla f = y^2\mathbf{i} + 2xy\mathbf{j} \text{ and } g(x, y) = b^2x^2 + a^2y^2 - a^2b^2 \Rightarrow \nabla g = 2b^2x\mathbf{i} + 2a^2y\mathbf{j}$$

So $\nabla f = \lambda \nabla g \Rightarrow y^2 = 2\lambda b^2x$ and $2xy = 2\lambda a^2y$

Multiplying the first equation by ay^2 and the second equation by b^2x , we get $a^2y^3 = 2\lambda a^2b^2xy = 2b^2x^2y$

We can exclude $y = 0$; it clearly cannot produce the maximum. Thus $a^2y^2 = 2b^2x^2$ and from $g(x, y) = 0$, $3b^2x^2 = a^2b^2$.

This gives us $x = \pm \frac{\sqrt{3}a}{3}$ and $y = \pm \frac{\sqrt{6}b}{3}$. The maximum occurs at $x = \frac{\sqrt{3}a}{3}, y = \pm \frac{\sqrt{6}b}{3}$; the value there is $\frac{2}{9}\sqrt{3}ab^2$.